

Hello Wednesday



Kickstart-kursus i programmering 23 dag 3

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Recap from Tuesday

- Python and Functions
- Animations
- Pair Programming
- Mouse and Key Input

Wednesday IFOs

- Variable Scoping
- Draw() function
- Conditionals
- Sounds
- Ideation

Variable Scoping Processing

- The principles of variable scoping in Processing.py largely follow those of Python but with some considerations given the environment of Processing's draw loop and function setup.
- Scoping in programming refers to the region or portion of the code where a variable or function is defined and can be accessed or modified.
- The concept of scoping is crucial for understanding variable lifetimes, visibility, and potential naming conflicts in your programs. It helps manage and organize data and functionality, enabling modular and maintainable code designs.

Variable Scoping Processing

Global Variables

Global Variables: Variables declared outside of any function are global to the sketch. They can be accessed and modified from any function, but if you want to modify them inside a function, you must declare them as `global` within that function.

```
var globalVarX = 50;

function setup() {
  createCanvas(400, 400);
}

function draw() {
  background(220);
  fill(255, 0, 0);
  ellipse(globalVarX, 100, 30, 30);
  globalVarX = globalVarX + 1;
}
```

Variable Scoping Processing

Local Variables

Local Variables: Variables declared inside a function are local to that function. They cannot be accessed outside of the function, and their memory is reclaimed once the function execution is complete.

```
function setup() {  
  createCanvas(400, 400);  
  var myVar = 42;  
  print(myVar); // This prints 42  
}  
  
function draw() {  
  background(220);  
  print(myVar); // This returns error, not defined  
}
```

Variable Scoping Processing

The setup() and draw() Functions: In Processing.py, `setup()` is called once at the beginning of the sketch, and `draw()` is called repeatedly, producing frames.

```
var x = 0;

function setup() {
  createCanvas(400, 400);
  var x;
  x = width / 2;
}

function draw() {
  background(220);
  ellipse(x, height / 2, 50, 50);
  x += 1;
}
```

Variables declared in `setup()` are local to `setup()`. Still, often you want to declare global variables at the top level of your sketch and then initialise or modify them in `setup()`.

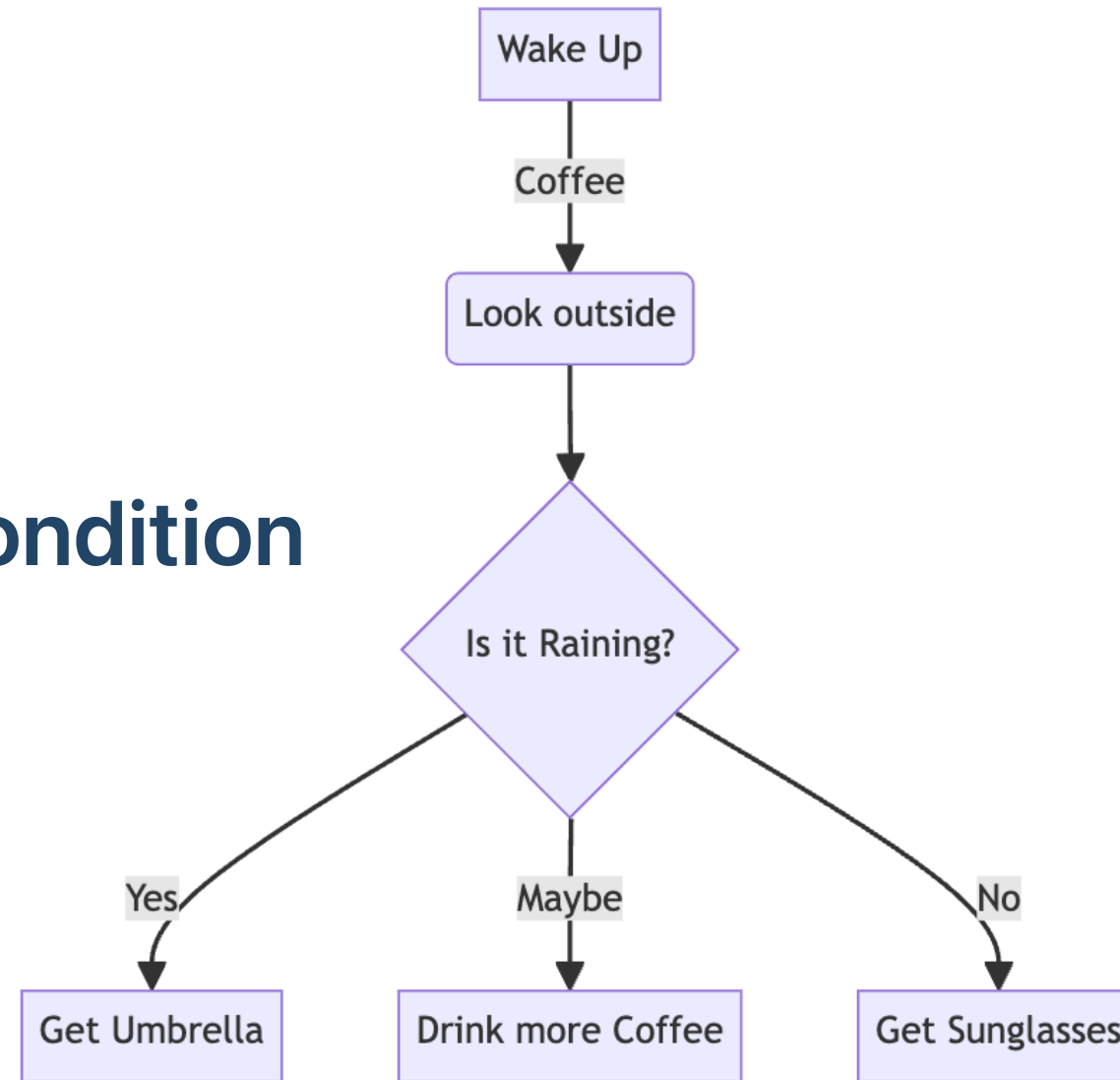
Conditionals in Computer Science

- Conditionals in computer science refer to constructs that allow for decision-making in code.
- They determine the flow of execution based on whether a given condition is true or false.
- Depending on the outcome of the condition, different blocks of code will be executed.
- Specifically, conditionals perform different computations or actions depending on whether a programmer-defined boolean condition evaluates to true or false.

Conditionals Conceptually

- **Basic Conditional (IF)**
 - **Concept:** If a specific condition is true, do something.
 - **Example:** "If it's raining, take an umbrella."
- **Alternative Path (ELSE)**
 - **Concept:** If the first condition isn't met, do something else instead.
 - **Example:** "If it's raining, take an umbrella. Otherwise, wear sunglasses."
- **Multiple Conditions (ELSE IF or ELIF)**
 - **Concept:** Check multiple conditions in sequence, and do the first thing that's true.
 - **Example:** "If it's raining, take an umbrella. If it's sunny, wear sunglasses. Otherwise, just go outside as usual."
- **Combining Conditions**
 - **Concept:** You can use logical operators (AND, OR, NOT) to combine conditions.
 - **Example:** "If it's a weekend AND the weather is good, go hiking."

Flowchart Condition



Conditionals

- **If Statement:** Executes a code block if a specified condition is true.
- **Else Statement:** Used in conjunction with an if statement, it specifies a block of code to be executed if the condition in the if statement is false.
- **Else If Statement:** Used to specify a new condition to test if the first condition is false.
- **Switch or Case Statement:** Allows a variable to be tested for equality against a list of values.

Conditionals in p5.js

Python supports the usual logical conditions from mathematics:

- Equals: `a == b` (Loose Equality Operator)
- Equals: `a === b` (Strict Equality Operator)
- Not Equals: `a != b`
- Less than: `a < b`
- Less than or equal to: `a <= b`
- Greater than: `a > b`
- Greater than or equal to: `a >= b`
- `||` checks multiple conditions and returns true if any one of the conditions is true.

Conditionals in p5.js I

if:

```
if (x > 10){  
  print("x is greater than 10")  
}
```

else:

```
if (x > 10) {  
  print("x is greater than 10");  
} else {  
  print("x is 10 or less");  
}
```

Conditionals in p5.js II

else if:

```
x = 13;  
if (x > 10) {  
  print("x is greater than 10");  
} else if (x == 10) {  
  print("x is 10 or less");  
}
```

You can also combine conditions using logical operators (and, or, not):

```
if (x > 10 && y < 5){  
  print("x is greater than 10");  
}
```

Conditionals in p5.js

For instance, to animate a circle moving across the screen and to make it wrap around when it reaches the edge:

```
var x_pos = 0; //set the x position

function setup() {
  createCanvas(400, 400);
}

function draw() {
  background(220);
  ellipse(x_pos, height / 2, 50, 50);
  x_pos += 2;
  if (x_pos > 400) {
    x_pos = 0; // resets x position
  }
}
```

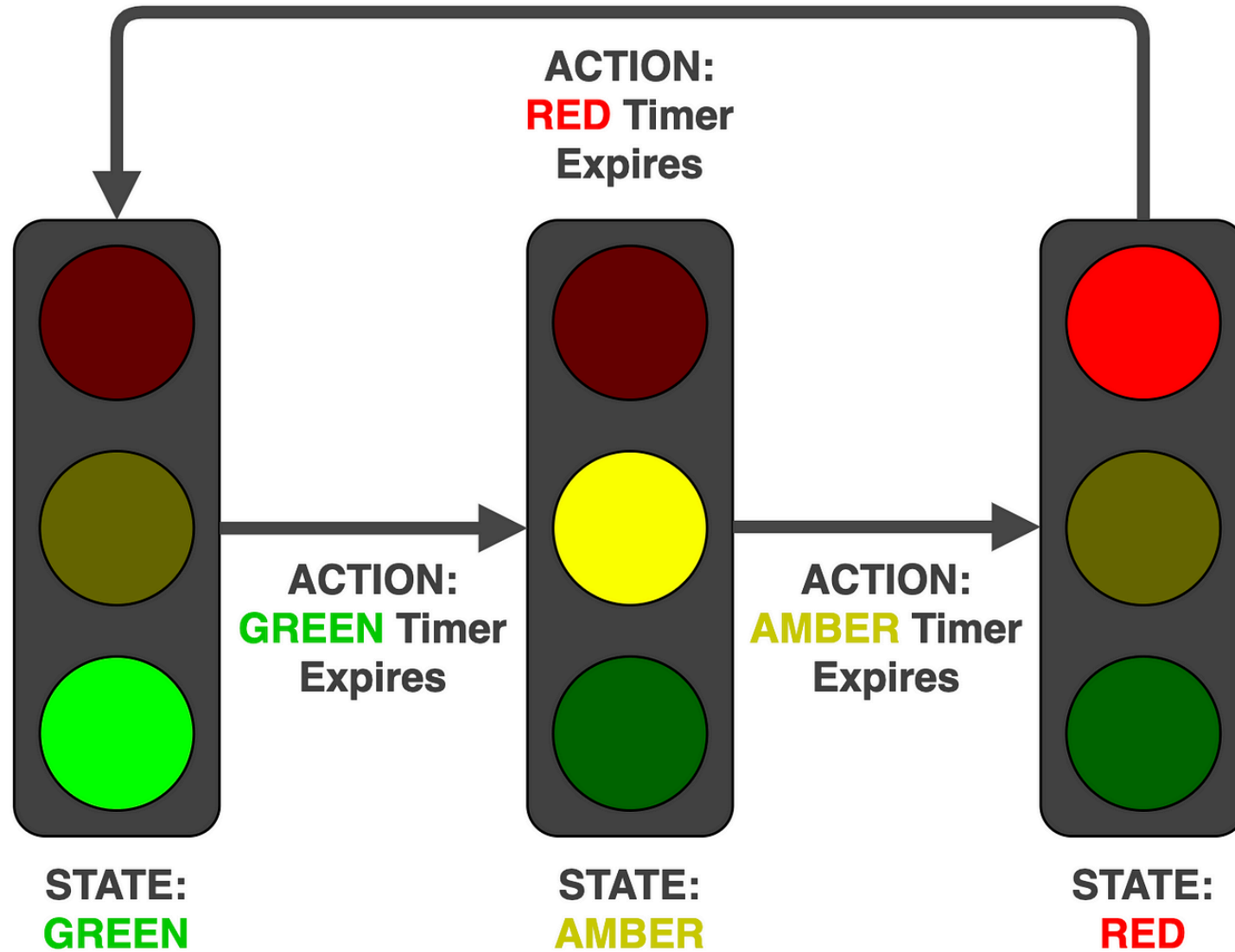

Finite State Machines- FSM

- A Finite State Machine (FSM) is a mathematical model of computation used to design algorithms.
- In the context of computer games, FSMs are often used for character behaviour, where different states might represent actions like "idle", "attack", "defend", or "flee", and game events or conditions determine transitions between states.

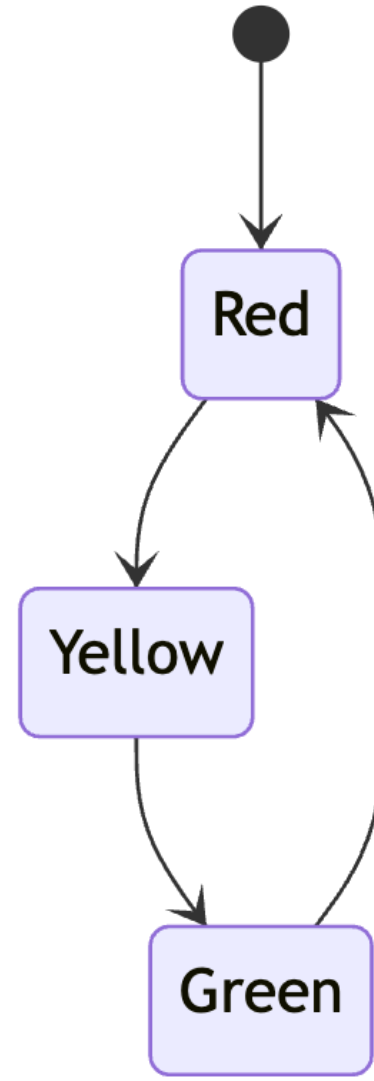
The STATES in a FSM

- **Discrete States:** An FSM consists of a limited or finite number of states. It can be in just one of these states at any given moment. Transitions define how it changes from one state to another based on inputs or conditions.
- **Transitions & Triggers:** Events or conditions trigger transitions between states. Each state specifies which state the machine will move to next for each possible input.
- **Start and End States:** Among the finite states, there is one initial state where the FSM begins its operation. Additionally, there can be one or more end states where the FSM is considered to be completed or final.

The Classic Example



State Diagram



Mathematical Abstraction of the FSM

- A finite state machine is a mathematical abstraction used to design algorithms. In simple terms, a state machine will read a series of inputs.
- When it reads an input, it will switch to a different state. Each state specifies which state to switch to for a given input.-

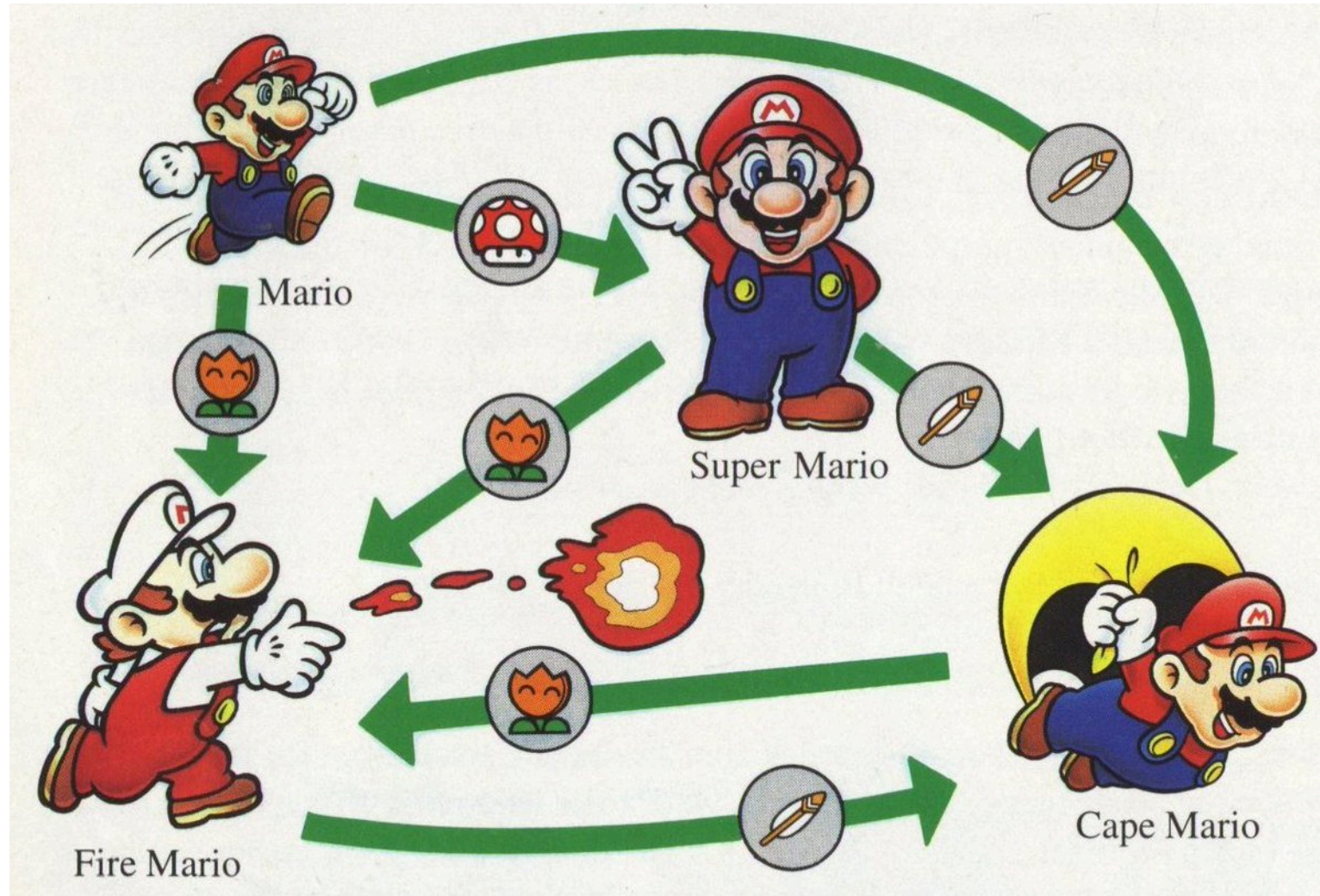
Game States

```
graph TD; STANDING[STANDING] -- "RELEASE (down arrow)" --> DUCKING[DUCKING]; STANDING -- "PRESS (down arrow)" --> DUCKING; STANDING -- "PRESS (B)" --> JUMPING[JUMPING]; JUMPING -- "PRESS (down arrow)" --> DIVING[DIVING];
```

A hand-drawn state machine diagram on grid paper showing four game states: STANDING, JUMPING, DUCKING, and DIVING. The transitions are as follows:

- STANDING** to **DUCKING**: Triggered by **RELEASE** (down arrow) or **PRESS** (down arrow).
- STANDING** to **JUMPING**: Triggered by **PRESS** (B).
- JUMPING** to **DIVING**: Triggered by **PRESS** (down arrow).

Mario States



FSM Code Example

```
//global vars
var state_on = "ON";
var state_off = "OFF";

//initial state
current_state = state_off;

function setup() {
  createCanvas(400, 400);
}

function draw() {
  background(220);

  if (current_state == state_on) {
    fill(0, 255, 0); // green is for on
  } else if (current_state == state_off) {
    fill(255, 0, 0); // red is for off
  }
  ellipse(width / 2, height / 2, 100, 100);
}

function mousePressed() {
  if (current_state == state_off) {
    current_state = state_on;
  } else if (current_state == state_on) {
    current_state = state_off;
  }
}
```

How do you add Yellow?

Sound in p5.js

- Let's keep it simple, but if interested explore the p5 reference
- We are going to load and play sounds, plenty other ways to work with sound.
- SoundFile object with a path to a file.
- The p5.SoundFile may not be available immediately because it loads the file information asynchronously.
- To do something with the sound as soon as it loads pass the name of a function as the second parameter.
- Only one file path is required. However, audio file formats (i.e. mp3, ogg, wav and m4a/aac) are not supported by all web browsers.

p5 sound demo

p5.js Web Editor | sound_01

https://editor.p5js.org/spikol/sketches/0jKN651VE

p5* File Edit Sketch Help English Hello, spikol!

Auto-refresh sound_01 by spikol

Sketch Files

- balltap.wav
- index.html
- JS sketch.js
- sound.mp3
- sound1.mp3
- style.css

sketch.js Saved: about 2 hours ago

```
1 let sound;
2
3 function preload() {
4   // Load the sound file
5   sound = loadSound('sound.mp3');
6   sound1 = loadSound('balltap.wav');
7 }
8
9 function setup() {
10  createCanvas(400, 400);
11 }
12
13 function draw() {
14  background(220);
15  textAlign(CENTER, CENTER);
16  textSize(24);
17  text('Press "S" or "A" to play sound', width / 2, height / 2);
18 }
19
20 function keyPressed() {
21  if (key === 's' || key === 'S') {
22    // Play the sound when 's' is pressed
23    sound.play();
24  }
25  if (key === 'a' || key === 'A') {
26    // Play the sound when 's' is pressed
27    sound1.play();
28  }
29 }
```

Preview

Press "S" or "A" to play sound

Console Clear

Code

```
function preload() {  
  // Load the sound file  
  sound = loadSound('sound.mp3');  
  sound1 = loadSound('balltap.wav');  
}  
  
function setup() {  
  createCanvas(400, 400);  
}  
  
function draw() {  
  background(220);  
  textAlign(CENTER, CENTER);  
  textSize(24);  
  text('Press "S" or "A" to play sound', width / 2, height / 2);  
}  
  
function keyPressed() {  
  if (key === 's' || key === 'S') {  
    // Play the sound when 's' is pressed  
    sound.play();  
  }  
  if (key === 'a' || key === 'A') {  
    // Play the sound when 's' is pressed  
    sound1.play();  
  }  
}
```